Lesson 1: Review: The Pythagorean Theorem

Naming the sides of a right triangle, then finding a missing hypotenuse or leg.

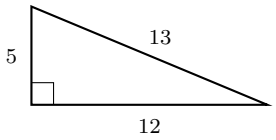
1. Right Triangles and Their Parts

A **right triangle** contains one 90° angle, marked with a small square. The two sides that meet to form that angle are the **legs**. The third side — the one *opposite* the right angle — is the **hypotenuse**, and it is always the **longest** side. A right triangle has two legs (in either order) but only ever one hypotenuse.

Example 1

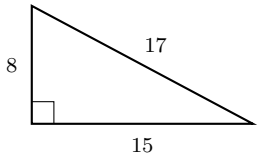
Name the two legs and the hypotenuse of each right triangle.

(a) Triangle A



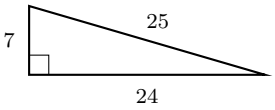
Legs: _____
Hypotenuse: _____

(b) Triangle B



Legs: _____
Hypotenuse: _____

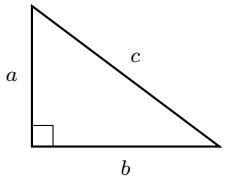
(c) Triangle C



Legs: _____
Hypotenuse: _____

2. The Pythagorean Theorem

Square the length of each leg and add the results: the total equals the square of the hypotenuse.

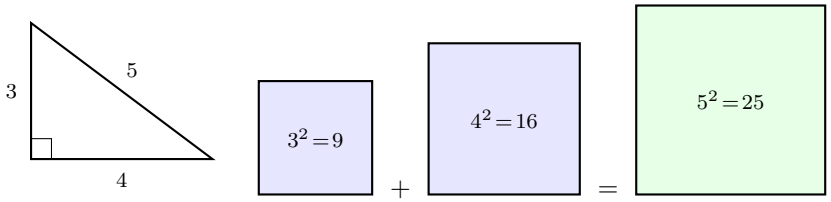


Pythagorean Theorem. In a right triangle with legs a and b and hypotenuse c,

$$a^2 + b^2 = c^2.$$

The letter c must be the hypotenuse — the side opposite the right angle, and the longest of the three — so c always sits alone on its own side of the equation.

Why is this true? Build a square on each side of a 3–4–5 right triangle. The areas of the two smaller squares add up exactly to the area of the largest square, and the same pattern holds for *every* right triangle, no matter its size.



Since $9 + 16 = 25$, the areas match: $a^2 + b^2 = c^2$.

Example 2

Verify the theorem by evaluating $a^2 + b^2$ and c^2 separately, then checking they are equal.

(a) Legs 3 and 4, hypotenuse 5.

(b) Legs 6 and 8, hypotenuse 10.

It works the other way too: if the two shorter sides give $a^2 + b^2 \neq c^2$, the triangle is **not** a right triangle.

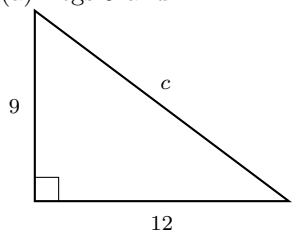
3. Finding the Hypotenuse

When both legs are known, solve for c by square-rooting both sides: $c = \sqrt{a^2 + b^2}$. Add the two squares *first*, then take the square root of the sum. If that sum is not a perfect square, simplify the radical as in Unit 1: $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$.

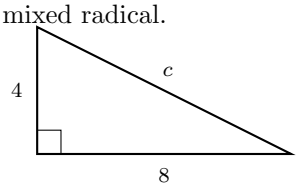
Example 3

Find the length of the hypotenuse in each right triangle.

(a) Legs 9 and 12.



(b) Legs 4 and 8. Give the exact answer as a simplified mixed radical.



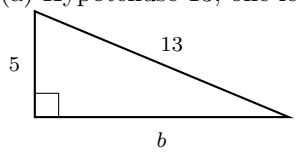
4. Finding a Missing Leg

If the hypotenuse and one leg are known, **subtract** instead of adding: $a = \sqrt{c^2 - b^2}$, or equivalently $b = \sqrt{c^2 - a^2}$. Always subtract the leg's square *from* the hypotenuse's square, never the reverse — the hypotenuse is the longest side.

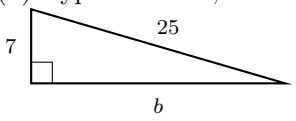
Example 4

Find the length of the missing leg in each right triangle.

(a) Hypotenuse 13, one leg 5.



(b) Hypotenuse 25, one leg 7.



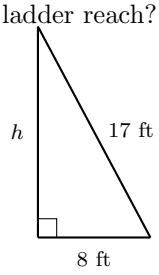
5. Real-World Applications

The theorem applies anywhere a right angle appears: a ladder against a wall, the diagonal of a rectangular field, a screen size. The first step never changes — find the right angle, decide whether the unknown side is the hypotenuse or a leg, then use the matching form of the theorem.

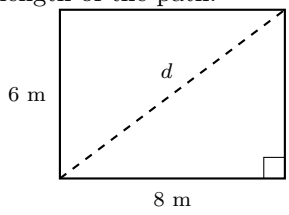
Example 5

Draw on the diagram, then solve.

(a) A 17 ft ladder leans against a vertical wall. Its base sits 8 ft from the wall. How high up the wall does the ladder reach?



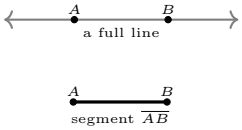
(b) A rectangular field is 6 m wide and 8 m long. A path is worn diagonally across it, corner to corner. Find the length of the path.



Lesson 2: Measuring the Length of a Line Segment

Finding the length of a horizontal or vertical segment by subtracting coordinates.

1. Lines, Segments, and the Cartesian Plane



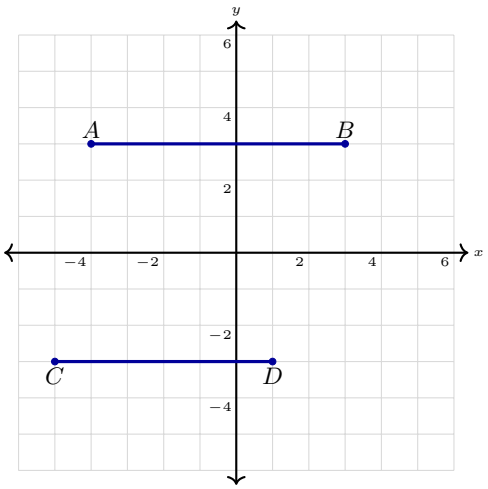
Every point on the **Cartesian plane** is named by an ordered pair (x, y): the x-coordinate gives the left/right move from the origin, the y-coordinate the up/down move. A **line** runs on forever in both directions, but a **line segment** is only the piece of a line between two **endpoints**, written segment AB or $\overline{AB}$. Because a segment stops at both ends, it has a definite, measurable **length**.

2. Length of a Horizontal Segment

A segment is **horizontal** when both endpoints share the same y-coordinate. Find its length by counting grid squares, or by subtracting the x-coordinates of the endpoints. Subtracting in the wrong order gives a negative, and a length is *always positive*, so take the **absolute value** — or simply subtract the smaller x from the larger: length $= |x_2 - x_1|$.

Example 1

Two horizontal segments are graphed on the grid below.



(a) State which coordinate (x or y) is the *same* for both endpoints of each segment. What does that tell you about the segment?

(b) State the coordinates of the endpoints of $\overline{AB}$ and of $\overline{CD}$.

(c) Find the length of each segment by counting grid squares.

(d) For each segment, subtract the x-coordinate of the left endpoint from the x-coordinate of the right endpoint. Compare your results to part (c).

(e) Complete the rule: for a horizontal segment, length $=$ |_____ − _____|.

Example 2

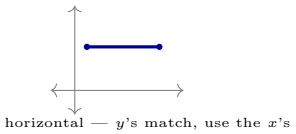
Determine each length. Show the subtraction you use.

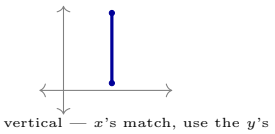
(a) Segment TU has endpoints $T(8, -3)$ and $U(-6, -3)$. Subtracting one way gives $8 - (-6) = 14$; the other way gives $-6 - 8 = -14$. Explain why both may be used for the length, and state the length of TU.

(b) Segment VW has endpoints $V(k - 4, m)$ and $W(k + 7, m)$. Determine the length of VW.

3. Length of a Vertical Segment

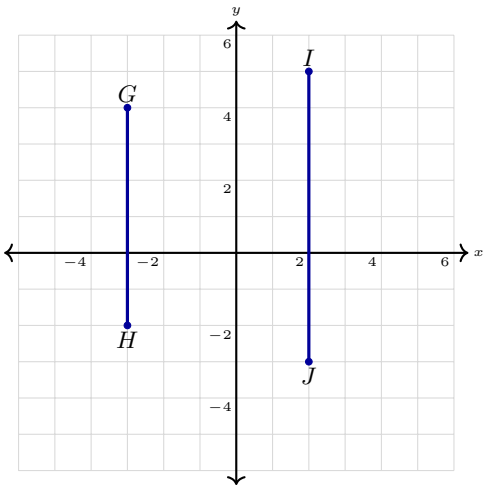
A segment is **vertical** when both endpoints share the same x-coordinate. The method is the same, turned 90°: subtract the y-coordinates and take the absolute value, since a length can never be negative — length $= |y_2 - y_1|$. Notice the swap: a horizontal segment shares a y and uses the x's; a vertical segment shares an x and uses the y's.





Example 3

Two vertical segments are graphed on the grid below.



(a) State which coordinate (x or y) is the *same* for both endpoints of each segment. What does that tell you about the segment?

(b) State the coordinates of the endpoints of $\overline{GH}$ and of $\overline{IJ}$.

(c) Find the length of each segment by counting grid squares.

(d) For each segment, subtract the y-coordinate of the lower endpoint from the y-coordinate of the upper endpoint. Compare your results to part (c).

(e) Complete the rule: for a vertical segment, length = |_____ − _____|.

Example 4

Determine each length.

(a) Segment KL has endpoints $K(5, -8)$ and $L(5, 6)$. Determine the length of KL.

(b) Segment MN has endpoints $M(h, n-5)$ and $N(h, n+9)$. Determine the length of MN.

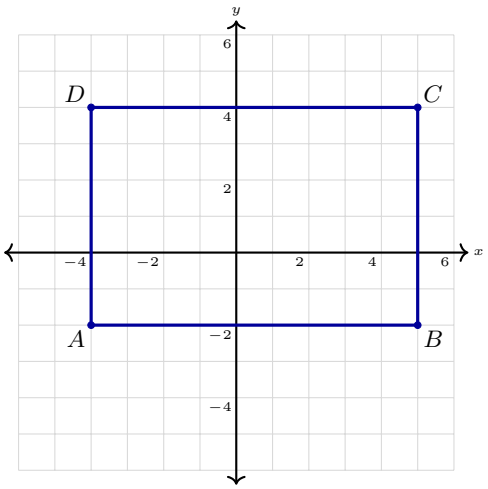
4. Applying Both Skills: Perimeter on a Grid

Shapes drawn on a grid often have only horizontal and vertical sides, so the **perimeter** (the distance around the outside) is just the sum of the side lengths found by the two rules above. In a rectangle opposite sides are equal, so Perimeter = 2(horizontal) + 2(vertical); for example, sides of 6 and 4 give $2(6) + 2(4) = 20$ units. A right triangle with a horizontal leg and a vertical leg has **area** $\frac{1}{2} \times$ (horizontal leg) $\times$ (vertical leg).

Working backwards. If you know the length and one endpoint, count that many units from the known endpoint — in either direction. A horizontal segment that starts at $(2, 1)$ and is 5 units long ends at $(7, 1)$ or at $(-3, 1)$.

Example 5

Rectangle $ABCD$ is graphed on the grid below.



(a) State the coordinates of all four vertices.

(b) Identify which sides are horizontal and which are vertical.

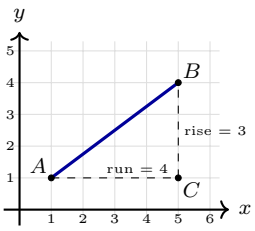
(c) Find the length of each of the four sides.

(d) Calculate the perimeter of rectangle $ABCD$.

Lesson 3: Measuring the Length of a Line Segment: Diagonal Segments and the Distance Formula

Turning any diagonal segment into a right triangle, then Pythagoras becomes the distance formula.

1. Turning a Diagonal Segment into a Right Triangle

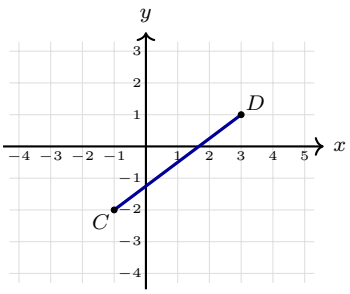


Most segments on a grid are neither horizontal nor vertical — they run **diagonally**, like AB at left, from $A(1,1)$ to $B(5,4)$. To measure one, build a right triangle underneath it: add a third point C sharing its x-coordinate with B and its y-coordinate with A, giving $C(5,1)$ and a right angle at C.
The horizontal leg AC is the **run** and the vertical leg CB is the **rise** — the same words used for slope. Each is found exactly as in Lesson 2: run $= 5-1 = 4$ and rise $= 4-1 = 3$. Segment AB is the hypotenuse, so

$$(AB)^2 = (\text{run})^2 + (\text{rise})^2 = 4^2 + 3^2 = 25 \quad \Longrightarrow \quad AB = \sqrt{25} = 5.$$

Example 1

The diagonal segment CD is graphed below, from $C(-1,-2)$ to $D(3,1)$.



(a) On the grid, draw the horizontal leg and the vertical leg needed to form a right triangle with hypotenuse CD. Label the right-angle vertex E.
(b) State the coordinates of the right-angle vertex E you drew.

(c) Find the length of the run (horizontal leg) and the rise (vertical leg).

(d) Use the Pythagorean theorem to calculate the exact length of CD.

2. Deriving the Distance Formula

Repeat the same construction with letters. For $P(x_1, y_1)$ and $Q(x_2, y_2)$, the helper point $R(x_2, y_1)$ gives run $= PR = x_2 - x_1$ and rise $= RQ = y_2 - y_1$, with the right angle at R. Since PQ is the hypotenuse, $(PQ)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$, and taking the square root gives the **distance formula**: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Because both differences are squared, it does not matter which point you call P and which you call Q — subtracting in the opposite order only flips the signs, and squaring erases them. A negative run or rise just means the segment goes left or down; squaring makes it positive. (A horizontal segment is the special case where the rise is 0, so the formula gives just the run.)

Example 2

Use the distance formula to determine the exact length of each segment. No grid needed — just substitute directly into $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

(a) $S(0,0)$ to $T(6,8)$

(b) $G(-3,4)$ to $H(1,1)$

3. Applying the Distance Formula Directly

With the formula in hand, no grid is required at all: label the two points (x_1, y_1) and (x_2, y_2), substitute, and simplify — including writing the resulting radical as a mixed radical in simplest form, exactly as in Unit 1.

Example 3

Use the distance formula to determine the exact length of each segment.

(a) $I(0,0)$ to $J(5,12)$

(b) $K(2,3)$ to $L(5,9)$

Example 4

Line segment MN has endpoints $M(-3,-2)$ and $N(3,0)$.

(a) Determine the exact length of MN, as a mixed radical in simplest form.

(b) Determine the length of MN as a decimal, rounded to the nearest tenth.

4. A Real-World Application

The formula is especially useful when a map or floor plan is placed on a coordinate grid: every location already has coordinates. Find the distance in grid units first, then multiply by the scale.

Example 5

A town places a coordinate grid over a park map. The **picnic shelter** is at $(3,8)$ and the **swimming dock** is at $(11,2)$, where each grid unit represents 25 m.

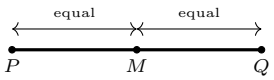
(a) Determine the distance between the picnic shelter and the swimming dock, in grid units.

(b) Use the map scale to convert your answer from part (a) into an actual distance, in metres.

Lesson 4: The Midpoint Formula

Averaging the coordinates of two endpoints to find the point exactly halfway between them.

1. What Is a Midpoint?



The **midpoint** M of a segment PQ is the one point exactly halfway between the two endpoints, so $PM = MQ$ and M splits the segment into two equal pieces. The distance formula $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ *subtracts* coordinates to measure how far apart two points are; the midpoint asks a different question — *where* is the point in the middle?

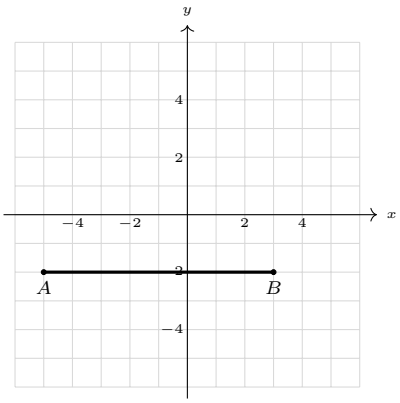
2. Locating a Midpoint by Counting

On a grid, the midpoint of a horizontal or vertical segment can be found just by counting squares: count the squares between the endpoints and step in half that many. For a diagonal segment, walk half the run across and half the rise up (or down).

Example 1

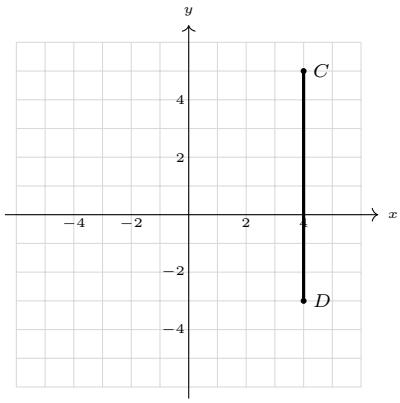
Count squares to locate and label the midpoint M, then record the three points.

(a) **Horizontal segment AB.** Describe how the x-coordinates of A and B combine to give the x-coordinate of M.



A(_____) B(_____) M(_____)

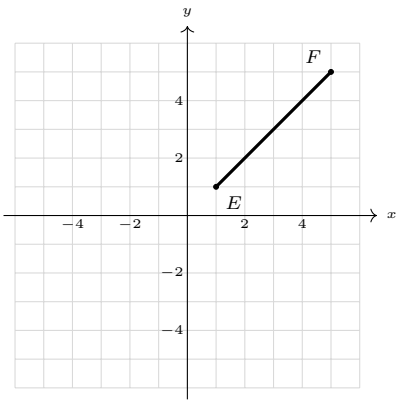
(b) **Vertical segment CD.** Describe how the y-coordinates of C and D combine to give the y-coordinate of M.



C(_____) D(_____) M(_____)

Example 2

The oblique segment EF below has $E(1,1)$ and $F(5,5)$.



Walk half the run and half the rise from E to locate the midpoint M, and label it on the grid. Record the coordinates of E, F, and M. Then check: does averaging the x-coordinates of E and F give the x-coordinate you found for M? Does averaging the y-coordinates give its y-coordinate?

3. The Midpoint Formula

The counting pattern becomes a formula. For $P(x_1, y_1)$ and $Q(x_2, y_2)$, the midpoint is $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ — **average the x's, average the y's**. Note the contrast with the distance formula, which *subtracts* the coordinates; here we *add* them and halve. With the formula there is no need for a grid at all.

Example 3

Determine the coordinates of the midpoint of the segment with each pair of endpoints.

(a) $P(2,8),\ Q(10,4)$

(b) $R(5,1),\ S(11,9)$

Example 4

Same task with negative coordinates — the formula is unchanged; add the signed numbers carefully.

(a) $A(-6,3),\ B(2,-7)$

(b) $C(-9,-4),\ D(3,-10)$

(c) $P(-3,4),\ Q(2,-1)$ (A midpoint does not have to land on a grid point.)

4. Midpoints on a Map

On a coordinate map, "exactly halfway between" means the midpoint of the segment joining the two places. Substitute the two locations into the formula just as with any other pair of points.

Example 5

A trail committee lays a coordinate grid over a map, with each unit representing 1 km. Fairview is at $(-15, 22)$ and Oakdale is at $(35, -8)$. The committee wants a rest area exactly halfway between the two towns. Determine its coordinates.

Lesson 5: Applications: Distance, Midpoint, and Finding a Missing Endpoint

Putting both formulas to work on maps and trails, and running the midpoint formula backward.

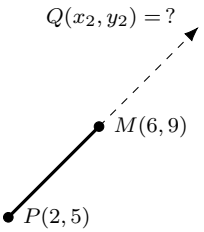
1. The Two Formulas

For $P(x_1, y_1)$ and $Q(x_2, y_2)$, the **distance** is $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ and the **midpoint** is $M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

Distance tells us *how far apart* two points are; the midpoint tells us *where halfway is*.

2. Finding a Missing Endpoint

If we know the midpoint M and only *one* endpoint, set the midpoint formula equal to the known midpoint and solve for the unknown coordinate — one small equation for x, one for y. Shortcut: $x_2 = 2x_M - x_1$ and $y_2 = 2y_M - y_1$, i.e. **double the midpoint, subtract the known endpoint**.



Worked demonstration. $P(2, 5)$ is an endpoint and $M(6, 9)$ is the midpoint. Solve each coordinate equation: $\frac{2 + x_2}{2} = 6 \Rightarrow 2 + x_2 = 12 \Rightarrow x_2 = 10$, and $\frac{5 + y_2}{2} = 9 \Rightarrow 5 + y_2 = 18 \Rightarrow y_2 = 13$, so $Q(10, 13)$. **Check** by putting both endpoints back in: $\left(\frac{2 + 10}{2}, \frac{5 + 13}{2}\right) = (6, 9)$. ✓

Example 1

$A(4, 1)$ is one endpoint of $\overline{AB}$ and the midpoint is $M(9, 6)$. Determine the other endpoint, B.

Example 2

Determine the coordinates of the missing endpoint in each case.

(a) $E(-6, 3)$ is one endpoint of $\overline{EF}$, with midpoint $M(-1, 7)$. Find F.

(b) $G(5, -8)$ is one endpoint of $\overline{GH}$, with midpoint $M(2, -3)$. Find H.

3. Real-World Distance Applications

When a map or floor plan is laid on a coordinate grid, the distance formula gives real distances straight from the coordinates. Compute the distance in *grid units* first, then multiply by the map scale to convert to metres or kilometres (1000 m = 1 km).

Example 3

A search-and-rescue team overlays a grid on a topographic map. The ranger station is at $(3, 4)$ and a hiker's radio signal is traced to $(8, 16)$, where each grid unit represents 200 m.

(a) Determine the distance between the ranger station and the hiker's signal, in grid units.

(b) Use the map scale to convert your answer from part (a) into an actual distance, in kilometres.

4. Real-World Midpoint Applications

The midpoint formula locates anything that must sit exactly halfway between two places — a meeting spot, a rest stop, the centre of a road or pipeline. The difference in coordinates from one endpoint to the midpoint tells how far east/west and north/south the halfway point lies.

Example 4

Meridian School at $(-6, 10)$ and Sunview School at $(18, -4)$ sit on opposite sides of town. The board wants a shared bus stop exactly halfway between them.

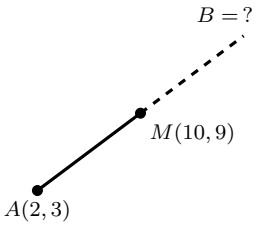
(a) Determine the coordinates of the bus stop.

(b) If each grid unit represents 100 m, how far east or west, and how far north or south, is the bus stop from Meridian School?

5. A Combined Application

Some problems need both tools chained together: run the midpoint formula in reverse to recover the missing endpoint, then use the distance formula on the result.

Example 5



An adventure park is designing a zip-line. The starting platform is at $A(2, 3)$, and the safety inspector marks the halfway cable anchor at $M(10, 9)$. Each grid unit represents 40 m.

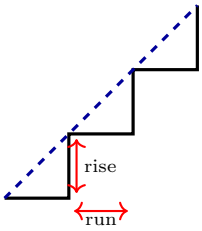
(a) Determine the coordinates of the ending platform, B, using the midpoint formula in reverse.

(b) Determine the total length of the zip-line, from A to B, in metres.

Lesson 6: Rise Over Run: The Slope of a Line Segment

Measuring steepness by comparing vertical change to horizontal change, then formalizing it.

1. What Slope Measures



Every step of a staircase goes up a little and across a little; a straight edge laid along the front corners is a line whose steepness depends on how the "up" compares with the "across." The **rise** is the vertical change between two points, the **run** is the horizontal change, and the **slope** m compares them:

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}.$$

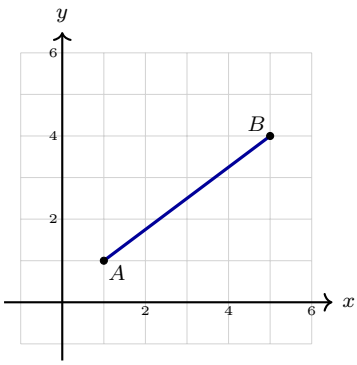
Rise is positive counting **up**, negative counting **down**; run is positive counting **right**, negative counting **left**. Every segment in this lesson's notes climbs to the right, so its slope is positive. (A segment that falls to the right has a negative slope; Lesson 7 looks at every kind of slope.)

2. Counting Rise and Run from a Graph

With both points plotted, just count squares: how many *up* from one point to the other's height, then how many *right* to the other's position. Do not read the two backwards — rise goes on top, run on the bottom.

Example 1

Points A and B are plotted on the grid below.



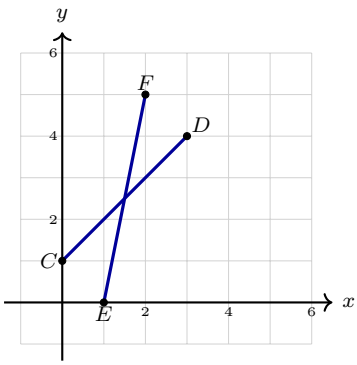
(a) Starting at A, count how many squares **up** you must go to reach the same height as B. This is the **rise**.

(b) Starting at A, count how many squares **right** you must go to reach the same horizontal position as B. This is the **run**.

(c) Use (a) and (b) to write the slope of $\overline{AB}$ as a fraction, $\frac{\text{rise}}{\text{run}}$.

Example 2

Count rise and run for each segment, then write its slope.



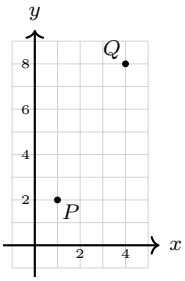
Segment	Rise	Run	Slope
CD			
EF			

3. The Slope Formula

When only the coordinates are given, the counting is done by subtraction. For $P(x_1, y_1)$ and $Q(x_2, y_2)$, $m = \frac{y_2 - y_1}{x_2 - x_1}$: the numerator is the rise, the denominator the run. Subtract *in the same order* top and bottom — whichever point you call (x_1, y_1), its coordinates are the ones subtracted, top and bottom — and reduce the fraction at the end.

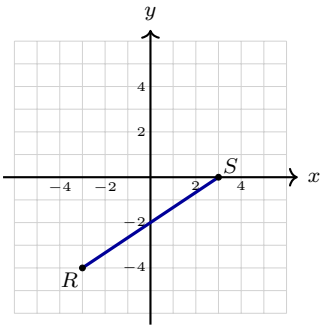
Example 3

Determine the slope of the line segment joining $P(1, 2)$ and $Q(4, 8)$.



Example 4

Determine the slope of the segment joining $R(-3, -4)$ and $S(3, 0)$. The negatives change nothing about the method — just track the signs, then simplify the fraction.

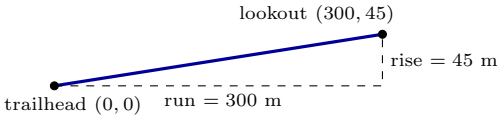


4. Slope in the Real World

Ramps, roads, and trails are described by the same ratio: vertical climb over horizontal distance. Read the rise and the run off the description, then reduce. A slope written as a percent is called the **grade**: for example, a slope of $\frac{1}{4} = 0.25$ is a 25% grade.

Example 5

A hiking trail climbs from the trailhead at $(0, 0)$ to a lookout at $(300, 45)$, each unit 1 m. Determine the slope of the trail as a fraction in lowest terms.



Lesson 7: Types of Slope: Positive, Negative, Zero, and Undefined

Recognising a line's slope type at a glance — and telling zero slope from undefined.

1. Positive Slope and Negative Slope

Slope compares rise to run: slope $= \dfrac{\text{rise}}{\text{run}} = \dfrac{\text{change in } y}{\text{change in } x}$. A line has **positive slope** if it **rises** from left to right — as x increases, y increases, so rise and run share the same sign and their quotient is positive. A line has **negative slope** if it **falls** from left to right — as x increases, y decreases, so rise and run have opposite signs and their quotient is negative.

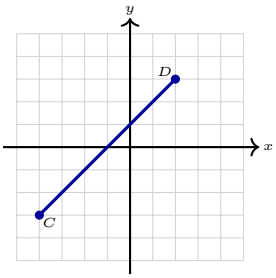
2. Zero Slope vs. Undefined Slope (the pair most often mixed up)

A **horizontal** line is flat: every point has the same y-coordinate, so the rise is always 0 and slope $= \frac{0}{\text{run}} = 0$. Zero is a real, perfectly ordinary number, and the line's equation is $y = k$. A **vertical** line runs straight up and down: every point has the same x-coordinate, so the *run* is always 0 and slope $= \frac{\text{rise}}{0}$. Dividing by 0 produces no number at all, so the slope is **undefined** — it does not exist; the line's equation is $x = k$. **Zero is an answer; undefined means there is no answer.** Flat like the horizon ⇒ slope 0; straight up like a flagpole ⇒ undefined.

Examples 1–4 — One Graph of Each Type

Work through all four graphs below; each shows a different type of slope.

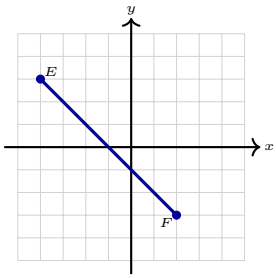
Example 1. Segment CD: $C(-4,-3)$, $D(2,3)$.



(a) Left to right, does it rise or fall? _____

(b) Calculate the slope of CD. Is it positive or negative, and does that match (a)?

Example 2. Segment EF: $E(-4,3)$, $F(2,-3)$ — same run as CD, but falling.

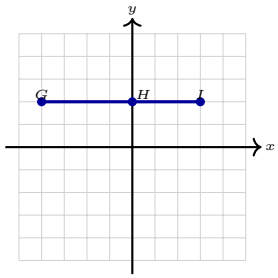


(a) Left to right, does it rise or fall? _____

(b) Calculate the slope of EF.

(c) Compare it to the slope of CD: what is the same, what is different, and why does that make sense?

Example 3. A horizontal line through $G(-4,2)$, $H(0,2)$, $I(3,2)$.

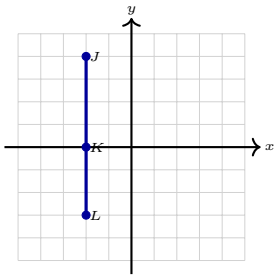


(a) Calculate the slope of GH, of HI, and of GI.

(b) All three agree — why must that happen for *any* two points on this line?

(c) Equation of the line: _____

Example 4. A vertical line through $J(-2,4)$, $K(-2,0)$, $L(-2,-3)$.



(a) Calculate the slope of JK, of KL, and of JL.

(b) All three agree — why must that happen for *any* two points on this line?

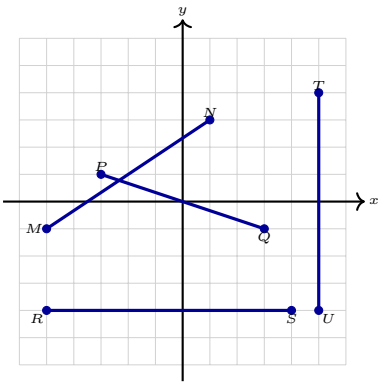
(c) Equation of the line: _____

3. Mixed Practice

On a test all four types arrive on one grid, so read each segment's tilt first, then confirm with the formula.

Example 5

Each segment has integer endpoints: $M(-5,-1)$ and $N(1,3)$; $P(-3,1)$ and $Q(3,-1)$; $R(-5,-4)$ and $S(4,-4)$; $T(5,4)$ and $U(5,-4)$.



(a) Complete the table: find the rise, run, and slope of each segment, then state its type (positive, negative, zero, or undefined).

Segment	Rise	Run	Slope	Type
MN				
PQ				
RS				
TU				

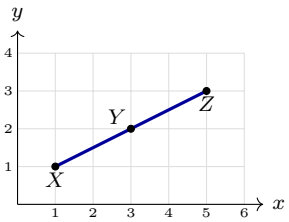
(b) One of these four segments has no slope number at all. Which one, and what is its equation?

Lesson 8: Determining Collinearity Using Slope

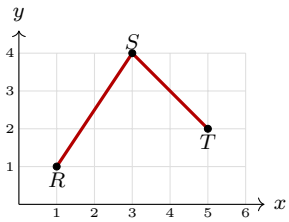
Three points lie on one straight line exactly when the two slopes through the shared middle point agree.

1. What "Collinear" Means

Three or more points are **collinear** if one single straight line passes through all of them at once — lay a ruler on the page, and its edge touches every point. If even one point sits off to the side of the line through the other two, the points are **not collinear**.



X, Y, Z are **collinear** — one straight segment covers all three.



R, S, T are **not collinear** — the path bends sharply at S.

2. The Slope Test

A picture only helps once the points are graphed. Given coordinates alone, use slope, $m = \dfrac{y_2 - y_1}{x_2 - x_1}$. **Key idea:** for points A, B, C with B shared by AB and BC, if $m_{AB} = m_{BC}$ the two segments point in exactly the same direction; since they also share the point B, they are two pieces of the *same* straight line, so A, B, C are **collinear**. If $m_{AB} \neq m_{BC}$, they bend at B and are **not** collinear. Any two segments that share a point work the same way, for example m_{AB} and m_{AC}.

Example 1

Determine whether $A(1, 2)$, $B(3, 6)$, and $C(6, 12)$ are collinear by comparing m_{AB} and m_{BC}.

$m_{AB} = \dfrac{y_B - y_A}{x_B - x_A} =$ $m_{BC} = \dfrac{y_C - y_B}{x_C - x_B} =$

Conclusion: _____

3. The Three-Step Process

Step 1: find the slope between the first two points. **Step 2:** find the slope between the second two points (sharing the middle point). **Step 3:** compare — equal slopes mean collinear, unequal slopes mean not collinear.

Example 2

Determine whether $D(-3, -4)$, $E(1, 0)$, and $F(5, 4)$ are collinear. Show both slopes, then conclude.

Step 1: $m_{DE} =$ Step 2: $m_{EF} =$ Step 3: compare. Collinear?

Example 3

Determine whether $G(-2, 5)$, $H(0, 1)$, and $I(3, -2)$ are collinear. Use the same three steps.

Step 1: $m_{GH} =$ Step 2: $m_{HI} =$ Step 3: compare. Collinear?

4. An Unknown Coordinate, and a Real-World Check

If we are *told* three points are collinear, the two slopes must already be equal. Set the known slope equal to the slope expression containing the unknown, then solve the resulting equation. The same test decides whether real locations on a map lie along one straight road.

Example 4

The points $J(2,3)$, $K(5,9)$, and $L(8,k)$ are collinear. Find k.

$m_{JK} = \dfrac{9-3}{5-2} =$ $\qquad$ $m_{KL} = \dfrac{k-9}{8-5} = m_{JK} \implies k =$

Example 5

On a town map the Fire Station is at $F(2,3)$, the School at $S(6,9)$, and the Library at $L(12,18)$, each unit 1 km. Do all three lie along one straight road?

$m_{FS} =$ $\qquad$ $m_{SL} =$

Conclusion: _____

Lesson 9: Parallel and Perpendicular Lines

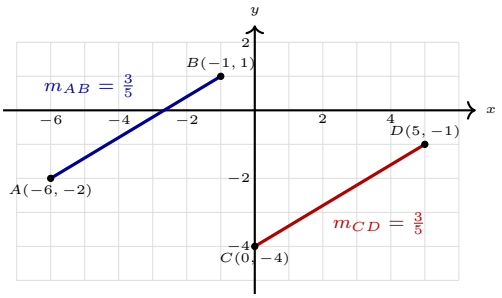
Comparing the slopes of two segments: equal slopes mean parallel, negative reciprocals mean perpendicular.

1. Slope, and the Two Relationships

For $P(x_1, y_1)$ and $Q(x_2, y_2)$, $m = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{\text{rise}}{\text{run}}$. Two segments are **parallel** exactly when their slopes are **equal** ($m_1 = m_2$), and **perpendicular** exactly when their slopes are **negative reciprocals** ($m_1 \cdot m_2 = -1$). A horizontal segment has slope 0 and a vertical segment has an *undefined* slope; those two are perpendicular to each other, but the product rule cannot be used on them.

2. Parallel Lines Have Equal Slopes

Parallel segments point in the same direction, so they must have the same steepness — and no matter how far you extend them, they never meet.



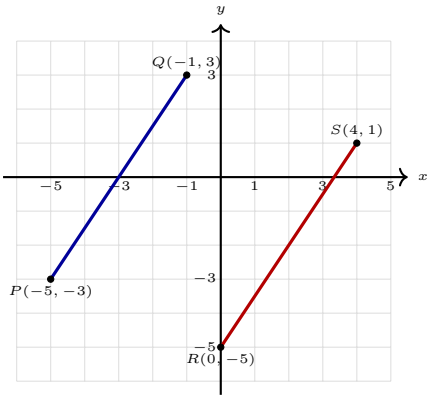
Worked demonstration. Segment AB (blue) and segment CD (red) sit on the same grid:

$$m_{AB} = \frac{1-(-2)}{-1-(-6)} = \frac{3}{5}, \qquad m_{CD} = \frac{-1-(-4)}{5-0} = \frac{3}{5}.$$

The slopes are identical, so $AB \parallel CD$. **Rule:** $AB \parallel CD$ if and only if $m_{AB} = m_{CD}$.
To *test* a pair, find each slope separately, then compare the two values.

Example 1

The graph shows segments PQ and RS.



(a) Calculate the slope of segment PQ.

(b) Calculate the slope of segment RS.

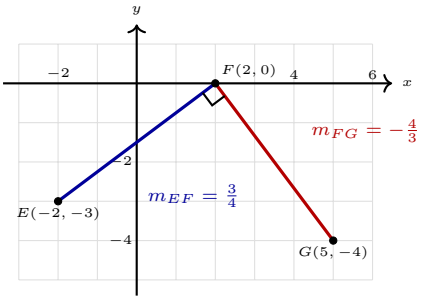
(c) Are PQ and RS parallel? Explain how you know.

Example 2

Segment TU has endpoints $T(-8, 5)$ and $U(4, -1)$; segment VW has endpoints $V(-3, -2)$ and $W(9, -8)$.
(a) Calculate the slope of TU. (b) Calculate the slope of VW. (c) Are TU and VW parallel? Explain.

3. Perpendicular Lines Have Negative Reciprocal Slopes

Perpendicular segments meet at a right angle (90°). To get the negative reciprocal of a slope, **flip it and change the sign**: the perpendicular of $\frac{5}{2}$ is $-\frac{2}{5}$, and the perpendicular of -3 (that is, $-\frac{3}{1}$) is $\frac{1}{3}$.



Worked demonstration. EF and FG meet at F, and the small square marks the right angle there:

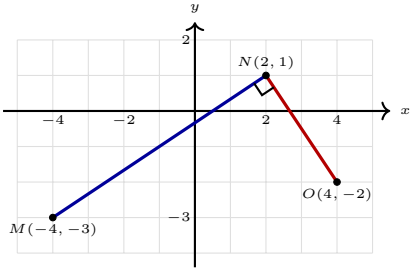
$$m_{EF} = \frac{0-(-3)}{2-(-2)} = \frac{3}{4}, \qquad m_{FG} = \frac{-4-0}{5-2} = -\frac{4}{3},$$

$$m_{EF} \times m_{FG} = \frac{3}{4} \times \left(-\frac{4}{3}\right) = -1.$$

Notice $-\frac{4}{3}$ is $\frac{3}{4}$ flipped, with the sign changed. **Rule:** two segments are perpendicular if and only if $m_1 \times m_2 = -1$. To test a pair, find both slopes and multiply.

Example 3

The graph shows segments MN and NO, which share the point N.



(a) Calculate the slope of segment MN.

(b) Calculate the slope of segment NO.

(c) Multiply the two slopes together. Are MN and NO perpendicular?

Example 4

Segment HI has endpoints $H(-7, 2)$ and $I(1, -2)$; segment JK has endpoints $J(-3, -6)$ and $K(1, 2)$.
(a) Calculate the slope of HI. (b) Calculate the slope of JK. (c) Multiply the slopes. Are HI and JK perpendicular?

Example 5

For each pair of slopes, state whether the segments are **parallel**, **perpendicular**, or **neither**.
(a) $m_1 = \frac{2}{5}$, $m_2 = \frac{2}{5}$ (b) $m_1 = -\frac{2}{7}$, $m_2 = \frac{7}{2}$ (c) $m_1 = \frac{5}{6}$, $m_2 = \frac{6}{5}$

Example 6

Two lines have slopes $-\frac{1}{2}$ and $\frac{k}{4}$. Determine the value of k if the lines are
(a) parallel (the slopes are equal). (b) perpendicular (the slopes multiply to -1).

Lesson 10: Unit Review: Coordinate Geometry

One reminder and a quick check for each of the nine ideas, then mixed challenges for the test.

Formula Summary

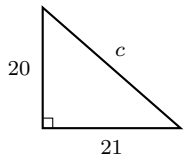
Everything in Unit 6 comes from these. Learn the table, then the questions are just careful substitution.

Pythagoras	$a^2 + b^2 = c^2$ (c = hypotenuse)	Slope	$m = \dfrac{\text{rise}}{\text{run}} = \dfrac{y_2 - y_1}{x_2 - x_1}$
Horizontal / vertical	length $= \lvert x_2 - x_1 \rvert$ / $\lvert y_2 - y_1 \rvert$	Types of slope	+ rises, − falls, 0 horizontal, undefined vertical
Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Collinear	$m_{AB} = m_{BC}$
Midpoint	$M\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$	Parallel	$m_1 = m_2$
Missing endpoint	$x_2 = 2x_M - x_1$, $y_2 = 2y_M - y_1$	Perpendicular	$m_1 \times m_2 = -1$ (negative reciprocals)

1. The Pythagorean Theorem

Legs a, b and hypotenuse c satisfy $a^2 + b^2 = c^2$: add the squares to find c, subtract to find a leg.

Example 1



(a) A right triangle has legs of length 20 and 21. Determine the length of the hypotenuse. (b) A right triangle has a hypotenuse of length 34 and one leg of length 16. Determine the length of the other leg.

2. Horizontal and Vertical Segment Lengths

Same y: horizontal, length $|x_2 - x_1|$. Same x: vertical, length $|y_2 - y_1|$.

Example 2

(a) Determine the length of the horizontal segment joining $E(-7, 4)$ and $F(5, 4)$.

(b) Determine the length of the vertical segment joining $G(3, 9)$ and $H(3, -6)$.

3. The Distance Formula

For a diagonal segment, the run and rise are the legs of a right triangle, so $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Example 3

(a) Determine the exact distance between $M(-2, -1)$ and $N(10, 4)$.

(b) Determine the exact distance, as a simplified mixed radical, between $P(-4, 1)$ and $Q(2, 5)$.

4. The Midpoint Formula

The midpoint is the average of the x's and the average of the y's: $M\left(\dfrac{x_1 + x_2}{2},\ \dfrac{y_1 + y_2}{2}\right)$.

Example 4

Determine the coordinates of the midpoint of the segment joining $A(-10, 3)$ and $B(6, -9)$.

5. Missing Endpoints and Applications

Know the midpoint and one endpoint? Run the formula backward — double the midpoint, subtract the known endpoint.

Example 5

(a) $M(5, -1)$ is the midpoint of segment $\overline{CD}$. If $C(-3, 7)$, determine the coordinates of D.

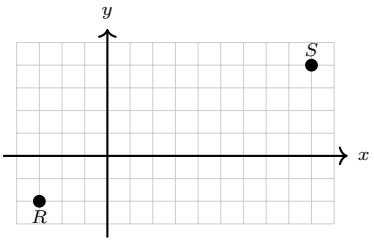
(b) Two trailheads are at $P(2, 9)$ and $Q(18, -5)$ on a park map, where each grid unit represents 10 m. A rest stop is built exactly halfway between them. Determine its coordinates, and state how far east and how far south it is from trailhead P.

6. Slope: Rise Over Run

$m = \dfrac{\text{rise}}{\text{run}} = \dfrac{y_2 - y_1}{x_2 - x_1}$ — subtract in the same order top and bottom.

Example 6

Determine the slope of the segment joining $R(-3, -2)$ and $S(9, 4)$.



7. Types of Slope

Rising ⇒ positive; falling ⇒ negative; horizontal ⇒ zero; vertical ⇒ undefined (zero and undefined are *not* the same).

Example 7

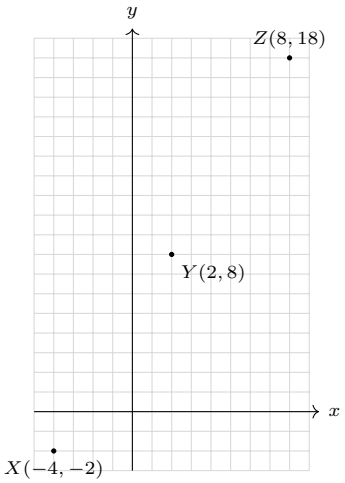
Classify the slope of each segment below as positive, negative, zero, or undefined.
(a) $A(-5,-3)$ and $B(4,3)$ (b) $C(-2,6)$ and $D(3,6)$ (c) $E(7,-4)$ and $F(7,5)$

8. Collinearity

Three points lie on one straight line exactly when the two slopes through the shared point match: $m_{AB} = m_{BC}$.

Example 8

Determine whether the points $X(-4,-2)$, $Y(2,8)$, and $Z(8,18)$ are collinear.



9. Parallel and Perpendicular Lines

Parallel $\Rightarrow$ equal slopes; perpendicular $\Rightarrow$ the slopes multiply to -1 (negative reciprocals).

Example 9

(a) Segment AB joins $A(-6,-1)$ and $B(2,5)$; segment CD joins $C(-3,4)$ and $D(5,10)$. Are AB and CD parallel, perpendicular, or neither?

(b) Segment EF joins $E(-2,3)$ and $F(4,-5)$; segment GH joins $G(1,-4)$ and $H(9,2)$. Are EF and GH perpendicular?

10. Mixed Review: Putting It All Together

Each question below chains several skills together — exactly what a test question looks like.

Challenge 1

Triangle PQR has vertices $P(-5,-3)$, $Q(3,1)$, and $R(-1,9)$.
(a) Determine the slope of PQ and the slope of QR, and show that the triangle has a right angle at Q.

(b) Determine the length of each side using the distance formula, and verify that the Pythagorean theorem holds.

(c) Determine the midpoint of the hypotenuse, PR.

Challenge 2

Quadrilateral $KLMN$ has vertices $K(-6,-2)$, $L(-2,4)$, $M(6,2)$, and $N(2,-4)$.

(a) Determine the slope of each of the four sides.

(b) Determine the length of each of the four sides.

(c) What is the most precise name for quadrilateral $KLMN$? Justify your answer.

Challenge 3

Segment JK joins $J(-4,-1)$ and $K(2,q)$. Segment LM joins $L(-3,5)$ and $M(3,-4)$.

(a) Determine the value of q so that $JK \parallel LM$.

(b) Using your value of q, determine the exact length of JK.